ASSIGNMENT#2 - REPORT

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2. **Menu (8 pts):** The program must have a menu that enables the user to select the required operation. Your program must do the operations between instruction 3 and instruction 8.

```
"C:\Program Files\Java\jdk-25\bin\java.exe" "-javaagent:C:\Program Files\JetBrains\IntelliJ IDEA 2025.2.3\lib\idea_rt.jar=52879" -Dfile.encoding=UTF-8 -Dsun.stdout.
Başlangıç verisi yüklendi.

--- PRODUCT STOCK MENU ---
1 - Yeni ürün ekle / güncelle
2 - In-order yazdır
3 - Miktar artır
4 - Miktar azalt
5 - Miktara göre artan yazdır
0 - Çıkış
Seçiminiz:
```

3. **Insert a new node (20):** If user selects this operation; she/he is prompted to enter barcode number, product name and product quantity properties. The product record is added to the tree sorted due to the barcode number. If you examine the sample illustration in Figure 2, all records are placed numerically ordered due to the barcode number.

```
Seçiminiz: 1
Barkod numarası: 252
Ürün adı: Spaceship
Miktar: 1
Ürün eklendi / güncellendi.
Değişiklikler inputBst.txt dosyasına kaydedildi.

--- PRODUCT STOCK MENU ---
1 - Yeni ürün ekle / güncelle
2 - In-order yazdır
3 - Miktar artır
4 - Miktar azalt
5 - Miktara göre artan yazdır
0 - Çıkış
Seçiminiz:
```

4. **Increase the quantity (15):** If user selects this operation, she/he is prompted to enter the barcode number and the increment amount. The quantity of the corresponding product is increased by the input amount and updated. The records are printed in order traversal.

```
Seçiminiz: 3
Barkod numarası: 252
Artış miktarı: 1
Miktar artırıldı. Yeni miktar: 2
Değişiklikler inputBst.txt dosyasına kaydedildi.
Güncel stok durumu (in-order):
12 | Milk | 24
37 | Bread | 30
45 | Eggs | 1210
58 | Butter | 10
63 | Cheese | 23
79 | Chicken Breast | 16
82 | Ground Beef | 1
94 | Apples | 100
101 | Bananas | 5
115 | Orange Juice | 15
128 | Rice | 2
133 | Pasta | 40
142 | Tomato Sauce | 8
156 | Olive Oil | 11
167 | Sugar | 18
173 | Salt | 45
184 | Coffee | 35
191 | Tea Bags | 20
196 | Laundry Detergent | 14
200 | Toilet Paper | 25
252 | Spaceship | 2
```

5. **Decrease the quantity(15):** If the user selects this operation, she/he is prompted to enter the barcode number, and the decrement amount. The quantity of the corresponding product is decreased by the input amount and updated. The records are printed in order traversal.

```
Seçiminiz: 4
Barkod numarası: 252
Azalış miktarı: 1
Miktar azaltıldı. Yeni miktar: 1
Değişiklikler inputBst.txt dosyasına kaydedildi.
Güncel stok durumu (in-order):
12 | Milk | 24
37 | Bread | 30
45 | Eggs | 1210
58 | Butter | 10
63 | Cheese | 23
79 | Chicken Breast | 16
82 | Ground Beef | 1
94 | Apples | 100
101 | Bananas | 5
115 | Orange Juice | 15
128 | Rice | 2
133 | Pasta | 40
142 | Tomato Sauce | 8
156 | Olive Oil | 11
167 | Sugar | 18
173 | Salt | 45
184 | Coffee | 35
191 | Tea Bags | 20
196 | Laundry Detergent | 14
200 | Toilet Paper | 25
252 | Spaceship | 1
```

6. **Print in-order (8 pts):**

```
Seçiminiz: 2
12 | Milk | 24
37 | Bread | 30
45 | Eggs | 1210
58 | Butter | 10
63 | Cheese | 23
79 | Chicken Breast | 16
82 | Ground Beef | 1
94 | Apples | 100
101 | Bananas | 5
115 | Orange Juice | 15
128 | Rice | 2
133 | Pasta | 40
142 | Tomato Sauce | 8
156 | Olive Oil | 11
167 | Sugar | 18
173 | Salt | 45
184 | Coffee | 35
191 | Tea Bags | 20
196 | Laundry Detergent | 14
200 | Toilet Paper | 25
252 | Spaceship | 1
```

7. **Print increasing due to the quantity (16 pts):**

```
--- PRODUCT STOCK MENU ---
1 - Yeni ürün ekle / güncelle
2 - İn-order yazdır
3 - Miktar artır
4 - Miktar azalt
5 - Miktara göre artan yazdır
0 - Çıkış
Seçiminiz: 5
Miktara göre artan sırada ürünler:
1 | 82 | Ground Beef
1 | 252 | Spaceship
2 | 128 | Rice
5 | 101 | Bananas
8 | 142 | Tomato Sauce
10 | 58 | Butter
11 | 156 | Olive Oil
14 | 196 | Laundry Detergent
15 | 115 | Orange Juice
16 | 79 | Chicken Breast
18 | 167 | Sugar
20 | 191 | Tea Bags
23 | 63 | Cheese
24 | 12 | Milk
25 | 200 | Toilet Paper
30 | 37 | Bread
35 | 184 | Coffee
40 | 133 | Pasta
45 | 173 | Salt
100 | 94 | Apples
1210 | 45 | Eggs
```